PORTFOLIO

BAYU AJI TRISASONGKO

Electrical Engineering Graduate | Mechanical Design, Electronics, Mechatronics & Robotics

Bringing mechanical design, electronics, and control together to turn engineering ideas into tangible prototypes. An electrical engineering graduate with a solid hands-on foundation in mechatronics, component fabrication, and rapid prototyping. Eager to contribute and learn within engineering teams developing robotics, automation, and precision mechanical systems.

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SCAN KODE QR ATAU KLIK DI MANA SAJA UNTUK BUKA PORTOFOLIO WEB
https://portobayu.vercel.app/
Bayu Aji Trisasongko · S1 Teknik Elektro, Universitas Telkom



Telkom University
S1 Teknik Elektro (2026)

Mechanical
3D CAD modeling, cycloidal gearbox design, CNC & 3D print fabrication.
`SolidWorks` `Cycloidal Reducer` `FDM 3D Print`

Electronics
Schematic design, circuit wiring, PCB layout, and power electronics.
`KiCad/Altium` `Relay Control` `Power Distribution`

Embedded Control
STM32 & Arduino MCU firmware, magnetic sensing, and closed loop.
`STM32 C/C++` `AS5600 Encoder` `PID Closed-Loop`

Prototyping
Physical mechatronics build, functional bench testing, and validation.
`Hardware Bench` `Rapid Iteration` `System Validation`

01 / PROJECTS

Engineering Work, in Practice · Selected Prototyping Showcase

Koleksi 5 proyek rekayasa fisik: Lengan Robot 5-DOF, Pendingin Surya, Robot Basket KRAI, Line Follower & Sumo Robot

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01 / FINAL PROJECT



Robot Manipulator 5-DOF ↗

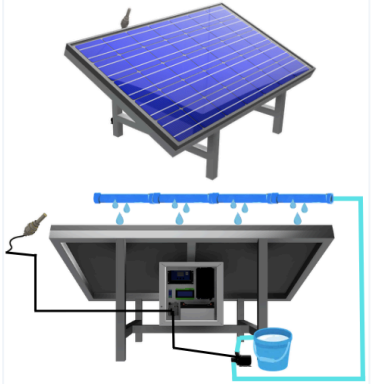
Prototipe lengan robot artikulasi 5-DOF bertenaga STM32F411 dengan gearbox cycloidal hasil 3D-print presisi. Mengintegrasikan encoder magnetik AS5600 closed-loop feedback untuk meminimalkan backlash pada sendi artikulasi gerak.

- Lengan 5-DOF & Reduksi Cycloidal
- STM32F411 MCU & AS5600 Feedback
- Akurasi Kinematika 79,8% (250–1.000 g)

STM32F411 MCU	NEMA 17 Stepper
TB6600 Driver	AS5600 Sensor
TCA9548A Mux	Cycloidal Gearbox

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02 / CAPSTONE DESIGN



Pendingin Panel Surya Otomatis ↗

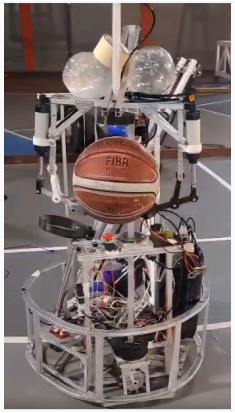
Sistem penyemprotan air otomatis untuk menurunkan temperatur operasional panel surya berbasis Arduino Uno. Memanfaatkan sensor suhu presisi dan relai pompa celup untuk memulihkan efisiensi konversi daya modul secara optimal.

- Sistem Semprot Air Otomatis Panel
- Dual Threshold Suhu (>30°C / <25°C)
- Pompa Celup 12V + LCD 16×2 Realtime

Arduino Uno R3	RTD Temp Sensor
5V Relay Module	12V Water Pump
16x2 I2C LCD	12V Battery Pack

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03 / KONTES ROBOT ABU INDONESIA (KRAI)



Kompetisi Robot Basket ↗

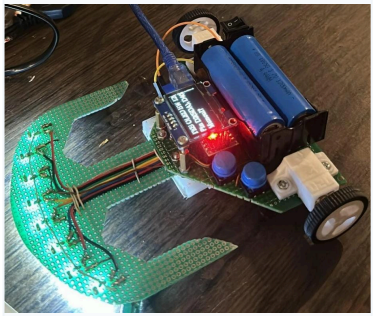
Robot penembak bola basket dinamis untuk Kontes Robot ABU Indonesia. Berperan aktif di tim mekanik merancang subsistem pelontar pneumatik 6-bar, mekanisme intake dribbler, serta chassis manuver roda omniwheel.

- Robot Penembak Basket Kompetisi KRAI
- Pelontar Pneumatik 6-Bar & Roda Omni
- Lolos Tahap Nasional ITB Jatinangor

6-Bar Pneumatics	24V Solenoid
RS775 DC Motor	Omniwheel Drive
Limit Switch & IR	Air Regulator

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04 / PROYEK MANDIRI



Robot Line Follower ↗

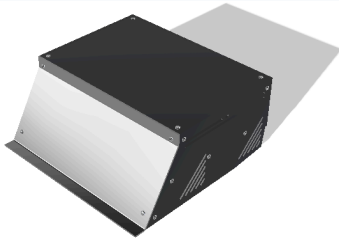
Robot pelacak garis otomatis berkecepatan tinggi dengan kemudi diferensial presisi. Dilengkapi bar sensor 8 pasang fotodioda LED, pemrosesan cepat Arduino Nano, serta pengemudi motor TB6612FNG responsif.

- Robot Pelacak Garis Kecepatan Tinggi
- Bar Sensor 8 Pasang LED & Fotodioda
- Motor N20 Metal Gear Diferensial

Photodiode Array	Arduino Nano
TB6612 Driver	N20 Gear Motor
0.96" I2C OLED	2S Li-ion Battery

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05 / PROYEK MANDIRI



Robot Sumo ↗

Robot sumo kelas kompetisi 1 kg berdimensi 20×20 cm dengan sasis akrilik kokoh dan bilah sekop aluminium. Mengandalkan 4 motor planetari PG24 torsi tinggi dan modul kendali nirkabel Bluetooth smartphone.

- Robot Sumo Kelas 1 kg Akrilik 20×20 cm
- 4× Motor Planetari PG24 (4 kg Force)
- Kendali Nirkabel Bluetooth Smartphone

Arduino Uno	4x PG24 Motors
L298N Driver	HC-05 Bluetooth
3S LiPo Battery	Acrylic Chassis

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5 Prototipe Rekayasa Fisik
Validasi hardware & fungsional 100%

Integrasi Mekatronika Lengkap
3D CAD, PCB, Firmware STM32 & Arduino

Pengalaman Kompetisi Robotik
Divisi Mekanik Nasional KRAI ITB

Dokumentasi & Video Lengkap
Model 3D interaktif di portobayu.vercel.app

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